

SAT Math Mastery · Week 1A

Linear Equations: Structure, Rearranging & Translating — Answer Key & Worked Solutions

Time	120 minutes: 90 min solving + 30 min self-check and error log
Topics	one-variable equations · solution-count structure · identities & coefficient matching · literal equations · word translation · inequalities
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Solve any linear equation (distribute, LCD for fractions/decimals, collect, divide).
S2. Spot a repeated *block* and solve for the block instead of the variable.
S3. Decide *one* / *none* / *infinitely many* solutions by comparing coefficients, before solving.
S4. Match coefficients when an equation is true for every x ; solve for constants.
S5. Rearrange literal equations (make any letter the subject).
S6. Translate words: “less than”, consecutive integers, ratios, fixed-plus-rate, ages.
S7. Linear inequalities: sign flip, compound statements, integer counts, rounding to a feasible whole number.
S8. Absolute-value equations: split into two cases and test each answer.

2 · REQUIRED FACTS

- Equal operations on both sides keep an equation true; multiplying or dividing an inequality by a negative number reverses it.
- “ n less than m ” = $m - n$; “at least” $\geq$; “at most” / “no more than” $\leq$; “more than” $>$.
- Consecutive integers differ by 1; consecutive *even or odd* integers differ by 2.
- A ratio $a : b = m : n$ means $a = mt$, $b = nt$; the whole is $(m + n)t$.
- $|u| = v$ means $u = v$ or $u = -v$, and requires $v \geq 0$.
- A sequence with a constant difference d has n th term $a_1 + (n - 1)d$.

3 · REQUIRED FORMULAS

$$ax + b = cx + d \implies (a - c)x = d - b, \quad x = \frac{d - b}{a - c} \quad (a \neq c)$$

$$P_{\text{rect}} = 2(\ell + w), \quad \text{total} = \text{fixed fee} + \text{rate} \times \text{quantity}$$

$$n, n + 2, n + 4, n + 6 \text{ (consecutive odd)}, \quad \text{sum of } k \text{ terms} = k \cdot (\text{mean})$$

4 · MINI THEORY SHEET

Master reduction. Every linear equation collapses to $mx = n$. Then:

$m \neq 0$	one solution	$x = \frac{n}{m}$
$m = 0, n \neq 0$	no solution	$0 = n$ is false
$m = 0, n = 0$	infinitely many	$0 = 0$ is true for all x

Identity in x . If two expressions are equal for every x , the coefficients of x match *and* the constants match.

Block rule. If an expression E occurs, $cE = k \Rightarrow E = \frac{k}{c}$; scale or shift the block, never expand it.

Inequalities. Same steps as equations; reverse only on $\times/\div$ by a negative. Compound: operate on all three parts.

Contexts. Fixed + rate: $y = \text{fee} + \text{rate} \cdot n$. For “at most” counts solve, then round down; for “at least” counts round up.

Absolute value. Solve $u = v$ and $u = -v$, then substitute each answer into the original equation.

5 · GUIDED EXAMPLES

Example 1 — Structure first (parameter)

For what value of k does $4(x - 2) + k = 4x - 3$ have infinitely many solutions?

Expand: $4x - 8 + k = 4x - 3$. The x -coefficients agree, so the constants must agree: $k - 8 = -3$, so $k = 5$.

Example 2 — Fractions: clear with the LCD

Solve $\frac{x+1}{2} - \frac{x-3}{5} = 2$.

LCD = 10: $5(x+1) - 2(x-3) = 20 \Rightarrow 3x + 11 = 20 \Rightarrow x = 3$.

Example 3 — Words $\rightarrow$ equation (consecutive integers)

The sum of three consecutive even integers is 54. What is the largest?

Let them be n , $n + 2$, $n + 4$: $3n + 6 = 54 \Rightarrow n = 16$. The integers are 16, 18, 20; the largest is 20.

6 · PRACTICE SET — 25 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

If $f(x) = 6(2x + 4)$, for what value of x is $f(x) = 48$?

(A) 8

(B) 2 ✓

(C) 4

(D) 6

Worked solution

1. Set $6(2x + 4) = 48$ and divide by 6: $2x + 4 = 8$.

2. Subtract 4: $2x = 4$.

3. Divide by 2: $x = 2$.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

What value of x satisfies the equation $\frac{2x + 5}{3} = x - 4$?

(A) -7

(B) 7

(C) 17 ✓

(D) 27

Worked solution

1. Multiply both sides by 3: $2x + 5 = 3x - 12$.

2. Subtract $2x$ and add 12: $17 = x$.

Answer: (C)

3. EASY · NO CALCULATOR NEEDED

The inequality $-5(x - 1) \leq 3(x + 3)$ is equivalent to which of the following?

(A) $x \geq -\frac{1}{2}$ ✓

(B) $x \geq \frac{1}{2}$

(C) $x \leq \frac{1}{2}$

(D) $x \leq -\frac{1}{2}$

Worked solution

1. Distribute: $-5x + 5 \leq 3x + 9$.

2. Subtract $3x$ and 5: $-8x \leq 4$.

3. Divide by -8 and reverse the inequality: $x \geq -\frac{1}{2}$.

Answer: (A)

4.

EASY · NO CALCULATOR NEEDED

If $p = \frac{5q-3}{6}$, which expression is equal to q ?

(A) $\frac{p}{5} + 3$

(B) $\frac{6p-3}{5}$

(C) $\frac{5p+3}{6}$

(D) $\frac{6p+3}{5}$ ✓

Worked solution1. Multiply by 6: $6p = 5q - 3$.2. Add 3: $6p + 3 = 5q$.3. Divide by 5: $q = \frac{6p+3}{5}$.**Answer: (D)**

5.

EASY · NO CALCULATOR NEEDED

$7x + 28 = 7x + k$. The equation has infinitely many solutions. What is the value of k ?

Worked solution1. The x -coefficients already match ($7 = 7$).2. Infinitely many solutions requires the constants to match: $k = 28$.**Answer: 28**

6.

EASY · NO CALCULATOR NEEDED

If $58 = 22 + 3(x - 5)$, what is the value of $3(x - 5)$?

Worked solution1. Treat $3(x - 5)$ as one block.2. Subtract 22 from both sides: $3(x - 5) = 58 - 22 = 36$.**Answer: 36**

7.

MEDIUM · NO CALCULATOR NEEDED

m , n , and p are three consecutive positive odd integers with $m < n < p$. Their sum is 63. What is the value of mp ?

Worked solution1. Let the integers be $n - 2$, n , $n + 2$. Their sum is $3n = 63$, so $n = 21$.2. Then $m = 19$ and $p = 23$.3. $mp = 19 \cdot 23 = 437$.**Answer: 437**

8.

MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\frac{3x+5}{4} = \frac{4x-1}{5}$?

Worked solution1. Cross-multiply: $5(3x + 5) = 4(4x - 1)$.2. Expand: $15x + 25 = 16x - 4$.3. Subtract $15x$ and add 4: $29 = x$.**Answer: 29**

9. MEDIUM · NO CALCULATOR NEEDED
If $\frac{3m}{4} + 2 = 8$, what is the value of $\frac{5}{2m}$?

Worked solution

1. Subtract 2: $\frac{3m}{4} = 6$.

2. Multiply by $\frac{4}{3}$: $m = 8$.

3. $\frac{5}{2m} = \frac{5}{16}$.

Answer: $\frac{5}{16}$

10. MEDIUM · NO CALCULATOR NEEDED
A woodworker charges a one-time fee of \$45 plus \$17 for each hour of work. The total charge for one job was \$164. For how many hours did the woodworker work?

- (A) 119 (B) 5
(C) 7 ✓ (D) 9

Worked solution

1. Write $45 + 17h = 164$.

2. Subtract 45: $17h = 119$.

3. Divide by 17: $h = 7$.

Answer: (C)

11. MEDIUM · CALCULATOR
A van can carry at most 3,600 kg. It is already loaded with 40 boxes of 25 kg each. Each additional crate weighs 18 kg. What is the greatest number of additional crates that can be loaded?

Worked solution

1. Current load: $40 \cdot 25 = 1000$ kg, leaving 2600 kg of capacity.

2. Inequality: $18c \leq 2600$, so $c \leq 144.4 \dots$

3. Only whole crates can be added, so round down: 144.

Answer: 144

12. MEDIUM · NO CALCULATOR NEEDED
Let a and b be numbers such that $20 < a < 30$ and $45 < b < 60$. Which of the following represents all possible values of $a - b$?

- (A) $-15 < a - b < 40$ (B) $-40 < a - b < -15$ ✓
(C) $15 < a - b < 40$ (D) $-40 < a - b < 15$

Worked solution

1. Smallest $a - b$: use the smallest a and the largest b : $20 - 60 = -40$.

2. Largest $a - b$: use the largest a and the smallest b : $30 - 45 = -15$.

3. So $-40 < a - b < -15$.

Answer: (B)

13.

MEDIUM · NO CALCULATOR NEEDED

The solution to $3x - k = 87$ is $x = 174$, where k is a constant. What is the value of k ?

Worked solution

1. Substitute $x = 174$: $3(174) - k = 87$.
2. $522 - k = 87$.
3. $k = 522 - 87 = 435$.

Answer: 435

14.

MEDIUM · NO CALCULATOR NEEDED

$N = \frac{Sp}{q-1}$, where N , S , and p are nonzero. Which expression is equal to q ?

(A) $\frac{Sp+1}{N}$

(B) $\frac{Sp}{N} - 1$

(C) $\frac{N}{Sp} + 1$

(D) $\frac{Sp}{N} + 1$ ✓

Worked solution

1. Multiply by $(q-1)$: $N(q-1) = Sp$.
2. Divide by N : $q-1 = \frac{Sp}{N}$.
3. Add 1: $q = \frac{Sp}{N} + 1$.

Answer: (D)

15.

MEDIUM · NO CALCULATOR NEEDED

Daniel is now five times as old as his sister Emma. In 4 years, Daniel will be three times as old as Emma will be then. How old is Daniel now?

Worked solution

1. Let Emma's age be e ; Daniel's is $5e$.
2. In 4 years: $5e + 4 = 3(e + 4)$.
3. $5e + 4 = 3e + 12 \Rightarrow 2e = 8 \Rightarrow e = 4$.
4. Daniel is $5e = 20$.

Answer: 20

16.

HARD · NO CALCULATOR NEEDED

$a(7x - 14) + 5a = 7(ax - 2a) - 35$. For what value of a does the equation have infinitely many solutions?

Worked solution

1. Left side: $7ax - 14a + 5a = 7ax - 9a$.
2. Right side: $7ax - 14a - 35$.
3. The x -terms match. Constants must match: $-9a = -14a - 35$.
4. $5a = -35$, so $a = -7$.

Answer: -7

17.

HARD · NO CALCULATOR NEEDED

If $a(x - 3) + 5 = bx - 2a + 11$ is true for all values of x , where a and b are constants, what is the value of a ?

Worked solution

1. Expand the left side: $ax - 3a + 5$.
2. Match x -coefficients: $a = b$.
3. Match constants: $-3a + 5 = -2a + 11$.
4. $-6 = a$.

Answer: -6

18.

HARD · NO CALCULATOR NEEDED

$$\frac{18x + 24}{6} - \frac{s}{11} = r(x - 9)$$

In the equation above, s and r are constants and $s > 0$. If the equation has infinitely many solutions, what is the value of s ?

Worked solution

1. Simplify the left side: $3x + 4 - \frac{s}{11}$. Right side: $rx - 9r$.
2. Match x -coefficients: $r = 3$.
3. Match constants: $4 - \frac{s}{11} = -27$.
4. $\frac{s}{11} = 31$, so $s = 341$.

Answer: 341

19.

HARD · NO CALCULATOR NEEDED

Let m and n be positive integers such that one-third of m is n less than one-half of m . Which of the following could be the value of m ?

(A) 12 ✓

(B) 16

(C) 8

(D) 10

Worked solution

1. Translate: $\frac{m}{3} = \frac{m}{2} - n$, so $n = \frac{m}{2} - \frac{m}{3} = \frac{m}{6}$.
2. Since n is an integer, m must be a multiple of 6.
3. Only 12 is a multiple of 6 (then $n = 2$).

Answer: (A)

20.

HARD · NO CALCULATOR NEEDED

For a certain number c , a sequence begins c , $2c + 3$, $3c + 6$, $4c + 9$, ... and the pattern continues. If the fifth term is 52, what is the tenth term?

Worked solution

1. The k th term is $kc + 3(k - 1)$.
2. Fifth term: $5c + 12 = 52 \Rightarrow c = 8$.
3. Tenth term: $10c + 27 = 80 + 27 = 107$.

Answer: 107

21.

HARD · NO CALCULATOR NEEDED

The smallest integer in a set of consecutive integers is -11 . The sum of the integers in the set is 12. How many integers are in the set?

Worked solution

1. The integers from -11 to 11 cancel in pairs and sum to 0.
2. The sum needs to be 12 more, so the next integer, 12, is included.
3. The set is $-11, -10, \dots, 12$: $12 - (-11) + 1 = 24$ integers.

Answer: 24

22.

HARD · NO CALCULATOR NEEDED

Maya is 4 years older than Noah. The square of Maya's age, in years, is 64 greater than the square of Noah's age. What is the sum, in years, of their ages?

Worked solution

1. Let Noah be N and Maya be $N + 4$.
2. $(N + 4)^2 - N^2 = 64 \Rightarrow 8N + 16 = 64$, so $N = 6$.
3. Maya is 10; the sum is 16.

Answer: 16

23.

HARD · DESMOS

What is the value of x that satisfies $|3x + 2| = 5x - 6$?

Worked solution

1. Case 1: $3x + 2 = 5x - 6 \Rightarrow x = 4$. Check: $|14| = 14$. ✓
2. Case 2: $3x + 2 = -(5x - 6) \Rightarrow 8x = 4 \Rightarrow x = \frac{1}{2}$. Check: $|3.5| = 3.5$ but $5(\frac{1}{2}) - 6 = -3.5$. Rejected.
3. The only solution is $x = 4$.

Answer: 4

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

For each integer $k \neq 1$, the equation $(k + 1)x + 3 = 2x + k$ has exactly one solution x . What is the sum of all integer values of k for which x is also an integer?

Worked solution

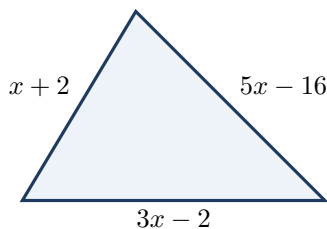
1. Collect: $(k - 1)x = k - 3$, so $x = \frac{k - 3}{k - 1} = 1 - \frac{2}{k - 1}$.
2. x is an integer exactly when $(k - 1)$ divides 2: $k - 1 \in \{\pm 1, \pm 2\}$.
3. So $k \in \{2, 0, 3, -1\}$, giving $x = -1, 3, 0, 2$.
4. Sum: $2 + 0 + 3 - 1 = 4$. ($k = 1$ gives $0 = -2$, no solution, so it is excluded.)

Answer: 4

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the triangle shown (not drawn to scale), the side lengths are $x + 2$, $3x - 2$, and $5x - 16$. The triangle is isosceles. What is the sum of all possible perimeters?

**Worked solution**

1. Three possible pairings of equal sides.
2. $x + 2 = 3x - 2$: $x = 2$, sides 4, 4, -6. Invalid (negative side).
3. $x + 2 = 5x - 16$: $x = \frac{9}{2}$, sides 6.5, 11.5, 6.5. Valid since $13 > 11.5$; perimeter 24.5.
4. $3x - 2 = 5x - 16$: $x = 7$, sides 9, 19, 19. Valid; perimeter 47.
5. Sum of possible perimeters: $24.5 + 47 = 71.5$.

Answer: 71.5